

AI in Clinical Practice — USMLE-Style Discussion Question Bank

Companion to the NIH/EU Unified Curriculum Blueprint

Course window: October 1, 2026 - March 2027

Each question is written in USMLE Step 2/3 vignette style (clinical stem → lettered options → answer → teaching explanation). They are meant to be projected or printed for small-group discussion at the end of each virtual class session, not for grading purposes — use them to surface disagreement and get students arguing about the “right” clinical action.

Master Schedule

Block	Dates	Curriculum Chapter	Theme
1	Oct 1 - Oct 28, 2026	Month 1	Admin Time Savings & AI Foundation
2	Oct 29 - Nov 25, 2026	Month 2	AI-Driven Diagnostics & Safety Override
3	Nov 26 - Dec 23, 2026	Month 3	Clinical Care Pathways & Data Sovereignty
4	Dec 24, 2026 - Jan 27, 2027	Month 4	Ethicolegal Autonomy & Final Checkoff
5	Jan 28 - Feb 24, 2027	Consolidation	Cross-Chapter Integration & Mock OSCE
6	Feb 25 - Mar 24, 2027	Capstone	Solo Pilot Simulation & Certification

Weekly cadence within each block mirrors the 3hr virtual / 3hr hospital-practice structure from the blueprint — drop one vignette into the last 15 minutes of each virtual session.

BLOCK 1 (Oct 1 - Oct 28, 2026): Admin Time Savings & AI Foundation

Competency targets: HITL workflow literacy, ambient scribing, prompt engineering (“Four Rs”)

Q1.1 — A 34-year-old family physician uses an ambient AI scribe during a visit for a

patient with chest pain radiating to the jaw. The AI-generated note documents “patient denies chest pain” based on a mistranscription. The physician signs the note without reading it and the patient later returns with an MI. What is the *most important* systemic failure illustrated here? A) The scribe’s speech-recognition accuracy B) Automation bias / failure of human-in-the-loop verification C) Inadequate malpractice insurance D) Lack of ICD-10 coding accuracy E) Patient’s own recall error

Answer: B. The core NIH/AAMC competency for Month 1 is recognizing that HITL means the human must actively verify AI output, not passively co-sign it. Speech errors (A) are expected and are exactly why review is mandatory; the sentinel failure is the clinician’s automation bias — trusting fluent-sounding text over verification.

Q1.2 — Which of the following prompt structures best exemplifies the AAFP “Four Rs” framework (Request, Reason, References, Revise) when asking an AI scribe to summarize a visit? A) “Summarize this.” B) “Summarize this visit for chronic pain follow-up, cite which patient statements support each assessment, and flag anything you are uncertain about for me to revise.” C) “Write a note that will pass insurance audit.” D) “Make this note sound more professional.” E) “Translate this transcript into German.”

Answer: B. Only B specifies a clear task (Request), asks for the rationale behind conclusions (Reason), asks the model to ground statements in the transcript (References), and invites clinician correction (Revise). The others are single-dimension prompts that omit the verification loop the framework exists to build.

Q1.3 — A German GP with limited technical background is being onboarded to an ambient scribe system. Per the curriculum’s “Germany Standard” for Month 1, what is the primary regulatory driver for requiring structured digital-literacy training rather than optional self-study? A) EU trade tariffs on software B) The documented shortage of technical AI literacy among German primary care physicians, addressed through structured onboarding C) US FDA device classification D) HIPAA training requirements E) ICD-11 rollout timelines

Answer: B. The blueprint frames Germany’s digitalization requirement around the documented literacy gap in general practice, not a US regulatory framework (HIPAA/FDA are irrelevant here).

Q1.4 — A resident says, “The scribe saved me two hours today, so I trust it completely now.” What teaching point from Month 1 should the attending raise? A) Time saved is inversely proportional to error rate B) Efficiency gains do not change the verification obligation — reduced documentation burden is a separate axis from output accuracy C) The resident should stop using the tool D) Only attendings should use ambient scribes E) Time savings validate the tool’s diagnostic accuracy

Answer: B. This is the “happiness factor vs. safety factor” tension the curriculum explicitly wants students to notice — administrative relief must not be mistaken for a trust signal about clinical accuracy.

■ BLOCK 2 (Oct 29 - Nov 25, 2026): AI-Driven Diagnostics & the Safety Override

Competency targets: critical appraisal of AI diagnostics, spotting hallucinations, German quality pillars (Functionality, Robustness, Explainability)

Q2.1 — A radiology CDSS flags a chest CT as “no acute findings” with 98% confidence. The supervising resident notices a subtle nodule the AI missed. What is the appropriate next action under a HITL safety framework? A) Trust the 98% confidence score and sign out the study B) Override the AI output, document the discrepancy, and route for independent radiologist review C) Increase the AI’s confidence threshold setting D) Ignore the finding since AI systems are validated pre-deployment E) Ask the AI to re-run the same model on the same image

Answer: B. High stated confidence is not evidence of correctness — this is the central “hallucination despite fluent/confident output” lesson of Month 2. The clinician’s job is exactly the manual override the blueprint’s sandbox exercises train.

Q2.2 — Under the German quality pillars framework (Functionality, Robustness, Comprehensibility/Explainability), a hospital CDSS gives a treatment recommendation but cannot show which patient variables drove the decision. Which pillar is violated? A) Functionality B) Robustness C) Comprehensibility (Explainable AI) D) Interoperability E) Scalability

Answer: C. Explainability under GDPR-aligned German standards requires that a clinician can trace *why* a recommendation was made — a black-box output without reasoning traceability specifically breaches Comprehensibility, distinct from whether it performs well (Functionality) or degrades under edge cases (Robustness).

Q2.3 — In a sandbox exercise, instructors seed a CDSS with a biased training-adjacent case (e.g., underrepresentation of a demographic subgroup), producing a systematically wrong risk score. What behavior should be *rewarded* per the Month 2 pedagogy? A) Accepting the AI score because it matches hospital protocol B) Detecting the anomaly, articulating why the score is implausible clinically, and executing a manual override C) Reporting the AI vendor to regulators immediately without internal review D) Disabling the CDSS permanently E) Waiting for IT to investigate before treating the patient

Answer: B. The blueprint explicitly designs this exercise to reward detection + override, not escalation-only or blind trust — patient care cannot wait on an IT ticket.

Q2.4 — A study shows an LLM answers USMLE-style board questions correctly at high rates but fails at multi-step bedside reasoning tasks. What does this dissociation best illustrate? A) The AI is unsafe for any clinical use B) Test-taking pattern-matching performance does not guarantee real-world complex clinical reasoning competence C) USMLE questions are poorly designed D) AI models

cannot process medical vocabulary E) Board scores are the best proxy for bedside safety

Answer: B. This is the specific “hallucination despite correct-looking answers” phenomenon the curriculum cites — strong benchmark performance and reliable clinical judgment are not the same construct.

■ BLOCK 3 (Nov 26 - Dec 23, 2026): Clinical Care Pathways & Data Sovereignty

Competency targets: AI-enhanced care planning, GDPR/data sovereignty, local/open-weight deployment rationale

Q3.1 — A German hospital wants to use a US-based cloud LLM API to generate nursing care plans from EHR notes. Legal counsel blocks this. What is the most likely GDPR-driven reason? A) The cloud vendor’s language model is outdated B) Cross-border transfer of identifiable patient data to a non-EU processor without adequate safeguards C) The hospital lacks sufficient internet bandwidth D) US models are technically incapable of German language processing E) The API does not support HL7 format

Answer: B. This is precisely why the blueprint states Germany “leans heavily toward private, local, open-weight AI deployments” — GDPR restricts moving identifiable health data outside the EU/adequate-safeguard zone, not a technical limitation of the model.

Q3.2 — A nurse uses an AI tool to draft a care plan combining NLP-extracted problems with a standardized nursing care model (e.g., NANDA-I). Before finalizing, what step is mandated by HITL principles? A) Auto-publish the plan to the EHR B) Clinician review and modification of AI-suggested nursing diagnoses against the actual patient presentation C) Skip review since NLP extraction is deterministic D) Have the AI vendor approve the plan E) Delay care until a physician re-types the entire note manually

Answer: B. Same HITL principle from Month 1, now applied to nursing workflows — automation should assist drafting, not replace clinical judgment.

Q3.3 — A researcher wants to use hospital AI-extracted data for a retrospective study. Per Month 3’s compliance content, what must be true of the dataset before external use? A) It must be raw and fully identifiable for accuracy B) It must be anonymized/pseudonymized and compliant with GDPR data-minimization principles C) It only needs verbal consent from the department head D) It must be stored on the researcher’s personal laptop E) No special handling is required if the AI generated it

Answer: B. GDPR compliant secondary use requires anonymization/pseudonymization and minimization — “researchers learn to extract anonymized, compliant data” is explicit in the blueprint.

Q3.4 — Why might an EU hospital prefer an on-premises, open-weight LLM over a proprietary cloud model for embedding into its EHR, according to the curriculum's data-sovereignty rationale? A) Open-weight models are always more accurate B) On-premises deployment keeps patient data within institutional/jurisdictional control, avoiding cross-border processing issues under EU law C) Cloud models cannot be updated D) Open-weight models require no maintenance E) Proprietary models are banned globally

Answer: B. Data sovereignty — not raw model performance — is the stated driver.

■ BLOCK 4 (Dec 24, 2026 - Jan 27, 2027): Ethicolegal Autonomy & the Final Checkoff

Competency targets: liability mapping, EU AI Act boundaries, informed consent, capstone readiness

Q4.1 — A patient asks whether an AI system was involved in generating their diagnostic report. Under EU AI Act transparency obligations and informed consent principles, what is the appropriate response? A) Deny AI involvement to avoid patient anxiety B) Disclose that AI-assisted tools were used, explain the clinician's role in verifying the output, and answer questions about the process C) Refuse to answer since it's proprietary D) Redirect the question to the software vendor only E) Say nothing unless the patient files a formal request

Answer: B. Transparency and informed consent are explicit Month 4 themes — patients are entitled to know AI was used and that a clinician remains accountable.

Q4.2 — During the "Solo Pilot" capstone simulation, a student runs a full shift (AI scribing → AI imaging validation → treatment planning) and the AI imaging tool disagrees with the student's own read of a fracture. What best demonstrates capstone-level competency? A) Automatically deferring to the AI since it passed validation testing B) Independently re-examining the image, reasoning through the discrepancy, and documenting the final clinical decision with justification C) Ordering a repeat scan to avoid making a decision D) Asking a classmate to decide E) Accepting whichever answer was reached first

Answer: B. The capstone specifically tests the "independent operator" competency — synthesizing AI input with independent clinical reasoning and defensible documentation, which also maps to liability mapping (the clinician, not the AI, bears documented responsibility).

Q4.3 — From a liability-mapping standpoint, if a clinician follows an AI recommendation that turns out to be wrong, who typically bears primary clinical responsibility under current EU/German medical liability frameworks discussed in this course? A) The AI vendor exclusively B) No one, since AI decisions are considered autonomous C) The treating clinician, who retains the duty to exercise independent clinical judgment regardless of AI input D) The hospital's IT department E) The patient, for consenting to AI-assisted care

Answer: C. This is the core ethicolegal point of Month 4 — AI is a decision-support tool, and clinical/legal responsibility remains with the human clinician, reinforcing why HITL was taught from Month 1 onward.

Q4.4 — Which combination correctly matches an EU AI Act risk-tiering concern to a clinical AI use case discussed in this course? A) Ambient scribing = prohibited AI practice B) Diagnostic CDSS influencing treatment decisions = high-risk AI system requiring human oversight, transparency, and robustness obligations C) Hospital Wi-Fi = high-risk AI system D) Care-plan NLP tools = banned under the AI Act E) All clinical AI is classified identically regardless of use case

Answer: B. Diagnostic/treatment-influencing systems are the paradigm “high-risk” category under the EU AI Act, which is why oversight, explainability, and robustness (Month 2’s pillars) all resurface here — the Act’s obligations scale with risk, they aren’t uniform or prohibitive by default.

■ BLOCK 5 (Jan 28 - Feb 24, 2027): Consolidation & Mock OSCE

Use these as integrative, cross-chapter stations.

Q5.1 — A student is completing a mock OSCE station: an ambient-scribed, AI-flagged “low risk” chest pain note for a patient whose vitals show new hypotension. Walk through, in order, every HITL checkpoint from Months 1–4 the student should apply before disposition. (*Open-discussion question — no single letter answer; rubric should require: verify transcription accuracy, question AI risk score against exam findings, invoke manual override per Month 2 training, confirm anonymization/documentation compliance per Month 3, document independent clinical reasoning and consent disclosure per Month 4.*)

Q5.2 — Compare and contrast “automation bias” (Month 1) with “explainability failure” (Month 2) using a single clinical example. (*Discussion prompt: automation bias is a human* failure to scrutinize; explainability failure is a system failure to justify. A good example shows both compounding — an opaque model output accepted without scrutiny.*)*

■ BLOCK 6 (Feb 25 - Mar 24, 2027): Solo Pilot Capstone & Certification

Full-shift simulation stations should draw one vignette from each prior block (4.1–4.4 style) plus a novel integration case combining scribing error + diagnostic override + data-sovereignty documentation + consent disclosure, scored against the Month 4 “Confidence Factor” checklist.

Notes for Instructors

- These vignettes are designed for discussion, not summative grading — expect legitimate disagreement, especially on Q2.1-Q2.3 and Q4.2, where reasonable clinicians may weigh override risk differently.
- Consider pairing each question with the actual EU AI Act risk-tier text and the German quality-pillars documentation referenced in the original curriculum blueprint for primary-source grounding.